

Vincent Zhou

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EDUCATION

University of Waterloo

Sept 2025 – June 2030

Bachelor of Applied Science, Computer Engineering, Honours Co-op

EXPERIENCE

Waterloo Aerial Robotics Group (WARG)

Autonomy Project Manager

Sept 2026 – Present

- Manage a **30-member** autonomy subteam, organizing contributors into parallel project groups, scoping the flight-software backlog into issues sized for onboarding, and tracking them through to flight readiness for the 2027 AEAC competition.
- Run code review for the autonomy stack and set its architectural direction, defining the ROS 2 node and message boundaries between the perception, GNC, and obstacle-avoidance packages so work split across groups integrates without rework.

Software Engineering Intern

May 2026 – Sept 2026

- Independently built and flight-tested a **visual teach-and-repeat** precision-landing service in **ROS 2** as concurrent action servers (Takeoff/Landing) over a multi-threaded executor with reentrant and mutually-exclusive callback groups, fusing camera, IMU, GPS, and rangefinder streams into a **10 Hz** control loop with cancellation support for pilot override.
- Designed the underlying algorithm: an altitude-keyed map of **ORB** features recorded during takeoff, matched on descent against the nearest stored frame via Lowe-ratio-filtered kNN, back-projected to metric ground coordinates, and resolved with **RANSAC** affine estimation, returning the drone to its launch point within **15 cm in flight test** and **2 cm in simulation**.
- Deployed the stack on an **NVIDIA Jetson** with a **CUDA** brute-force matcher (CPU fallback) inside a **10 Hz** vision/control loop; architected the non-optimal teach-and-repeat pipeline as a **ROS 2/py_trees** mission (Takeoff → FlyAround → RTL → Landing) with altitude-keyed ORB map recording, nearest-altitude frame matching on descent, PI body-velocity setpoints over **MAVROS**, and handoff to **ArduPilot**'s LAND mode with debounced ground confirmation.
- Engineered the failure path with a stale-vision recovery state machine, inlier-ratio and scale validation to reject degenerate matches, and a debounced handoff to **ArduPilot**'s LAND mode so the autopilot owns touchdown.
- Architected a full-stack, real-time **UAV geolocation platform** (React/TypeScript frontend, **Flask** REST API) for the 2026 AEAC Firefighting Competition, enabling ground operators to capture, annotate, and geolocate targets spotted mid-flight using a pinhole camera model and ray-plane intersection against live telemetry to localize targets in 3D to **<0.5 m**.

PROJECTS

Classly | Classroom Management Platform: React 19, Node.js/Express 5, Postgres

- Built a full-stack classroom platform (classes, invites, enrollments, weighted grades) on **Postgres/Neon + Drizzle**, with role-based access (admin/teacher/student) and **Better Auth** sessions (email/password + Google/GitHub OAuth) behind a Vercel api proxy for first-party cookies.
- Migrated the API from a long-running **Render** service onto **Cloudflare Workers**, cutting idle cold starts from 10–30s to near-instant; kept Express via Node compatibility, used Neon HTTP for reads/writes and request-scoped WebSocket pools for transactions, and added cron-based session cleanup with secrets managed via **Wrangler**.
- Hardened concurrent enrollment/grade writes with pessimistic transactional locking, and secured the API with **Arcjet** bot detection/rate limiting and parameterized queries.

SITL-Plus | UAV Software-in-the-Loop Simulator, WARG

- Built a from-scratch simulator using **PyBullet** rigid-body physics to model quadcopter flight dynamics, bridged to **ArduPilot**'s SITL binary over a **UDP JSON** protocol for high-rate physics/flight-controller state exchange.
- Simulated onboard sensors: downward and forward RGB-D cameras via the OpenGL renderer and a ray-cast rangefinder/LiDAR, streamed to **Rerun** over **gRPC** for live 3D visualization, enabling integration testing of the vision and landing pipelines without hardware.
- Containerized the stack (physics engine + ArduPilot SITL/MAVProxy) with **Docker Compose**, covered the sensor and dynamics models with a **pytest** suite, and used the simulator to validate and tune the precision-landing pipeline to **2 cm** before committing to flight tests.

MacroMe | Autonomous Web Agent

2nd Place, WebAgent Track

Battle of the Schools (UTMIST × Wat.ai)

- Built an **autonomous web agent** in **TypeScript/Node.js** using **Steel cloud browsers** and **Playwright** to navigate DoorDash, compare menus, and assemble meal carts through **LLM-guided selection**, with human approval before checkout.
- Integrated **SQLite FTS5** retrieval over USDA nutrition data to inform meal ranking, and built a **custom MCP server** exposing daily macro summaries, saved digests, and email delivery to AI assistants.

Health Pro Wellness Clinic | Astro, Cloudflare Pages

- Shipped a production trilingual (EN / Simplified / Traditional Chinese) static site for a local clinic on **Cloudflare Pages**, with a CI audit harness enforcing WCAG AA contrast, a hash-based **CSP**, transfer-weight budgets, and layout regression across 11 viewports via headless Chrome.

SKILLS

- **Languages:** Python, C++, TypeScript/JavaScript, Java, SQL, Bash, Lua, Verilog
- **Robotics & Vision:** ROS 2, ArduPilot/MAVROS, MAVLink, OpenCV, CUDA, PyBullet, Rerun, NumPy, Eigen, NVIDIA Jetson, Raspberry Pi, PyTorch
- **Backend & Agents:** FastAPI, Flask, Node.js/Express, REST, PostgreSQL, Drizzle ORM, Neon, Pydantic, Zod, Playwright, CDP, Steel, MCP, PyTorch, TCP/UDP
- **Infrastructure & Testing:** Docker, Cloudflare Workers/Pages, Wrangler, Git/GitHub, GitHub Actions CI, Linux, pytest, Vitest